Resolution refinement followed by the rodeo algorithm for eigenstate preparation: lattice refinement of the Hubbard model

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We extend the resolution-refinement-plus-rodeo pipeline that we implemented and assessed on the pairing Hamiltonian [1] to the Hubbard model, the standard testbed for strongly correlated fermions, and to the form of refinement for which the method was originally proposed: a *lattice* refinement in which every coarse site is resolved into a block of fine sites and the prolongation maps each coarse orbital onto the bonding combination of its block. The Galerkin projection of the fine Hubbard Hamiltonian onto the prolonged subspace is again a Hubbard model: in one dimension with $(t, U) \to (t/2, U/2)$, so that the refinement hierarchy is self-similar at fixed U/t, and in two dimensions with $(t, U) \to (t/2, U/4)$. All stages — the coarse eigenstate, the prolonged state, the adiabatically refined state and the rodeo-filtered state — are benchmarked against exact diagonalization in the conserved $(N_\uparrow, N_\downarrow)$ sector, using the dense, Lanczos and matrix-free solvers of our ED code, and the same matched-observable methodology as in Ref. [1]. For the open chain refined from four to eight sites with two particles per spin, the prolonged state has a ground-state weight $p_0 = 0.75$–0.81 for $U/t = 0$–8 and the shift-controlled refinement path brings it to $p > 0.9999$ by $T = 16/t$ at every U, although the sector gap of the fine Hamiltonian falls from $0.53t$ to $0.14t$: the path conserves total spin and lattice symmetry, so the gap that controls the refinement is the distance to the next state in the symmetry sector of the target, $0.88t$–$0.37t$ here, not the sector gap. The same protection lets the refinement $8 \to 16$ sites succeed at $T = 16/t$ with a fine-sector gap of $0.05t$. The lesson has a sharp converse: for the $2\times2 \to 4\times4$ plaquette refinement at $(N_\uparrow, N_\downarrow) = (2, 2)$ the coarse ground state is a d-wave singlet and has *zero* overlap with the s-wave fine ground state; refining the nearly degenerate s-wave coarse state instead gives $p = 0.99997$ at $T = 16/t$. A two-local alternative to the projector path, adiabatic "un-dimerization" of the fine lattice, refines as fast but is capped by the finite-penalty ceiling $1 - p \simeq (t/\Lambda)^2$; two rodeo cycles remove either residual. Chains that start from the filled dimer, a single computational basis state, reach $p > 0.999$ in two refinement steps without any variational input. [Circuit-level numbers (Trotterized refinement, controlled evolution, CNOT budgets), the UCCSD comparator and the strong-coupling (t–J) limit are to be added.]

I. INTRODUCTION

In Ref. [1] we implemented, end to end and at the gate level, a pipeline for eigenstate preparation in which an eigenstate of a coarse Hamiltonian is lifted by a prolongation operator into a finer model space, transported adiabatically along a shift-controlled, gapped path (*resolution refinement* [2]) and then cleaned by the rodeo algorithm [3, 4], a projective spectral filter. The testbed there was the constant-pairing Hamiltonian, exactly solvable by the Richardson ansatz, with "resolution" meaning the number of single-particle levels at fixed particle number, and the assessment was sober: at eight qubits the refinement stage did not beat a cheap unitary coupled-cluster input on gate count, and its merits were the absence of a classical optimization loop and of training measurements. A seniority-breaking particle-hole interaction, added in Sec. VI of that work, left the structure and convergence of the pipeline unchanged while raising the depth of a Trotter step ninefold.

The present paper carries the same pipeline, the same benchmarks and the same cost accounting to the Hubbard model [5, 6],

$$H = -\sum_{\langle ij\rangle\sigma} t_{ij}\left(c^\dagger_{i\sigma}c_{j\sigma} + \text{h.c.}\right) + U\sum_i n_{i\uparrow}n_{i\downarrow}, \qquad (1)$$

for three reasons. First, the Hubbard model is the canonical strongly correlated fermion problem and the most frequently used benchmark for quantum simulation of

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fermions [7–10]; it contains the one-body particle-hole channel (hopping) that the pairing model lacks and the two-body on-site term in their simplest forms, so it tests the lesson of Sec. VI of Ref. [1] on a model where the seniority symmetry is absent from the outset. Second, it is the setting for which resolution refinement was proposed [2]: a *lattice* refinement in which every coarse site is resolved into a block of fine sites, the prolongation is a product of Givens rotations rather than the gate-free embedding of the basis refinement, and the fine lattice is the natural discretization of a continuum problem. Third, the model is not exactly solvable in general, so the exact-diagonalization (ED) reference that anchored every number in Ref. [1] must now be produced numerically; we keep that discipline and benchmark every stage against ED in the conserved $(N_\uparrow, N_\downarrow)$ sector with the dense, Lanczos and matrix-free solvers of our ED code [12], which also covers the t–J model and the chain, square and checkerboard lattices used below.

Two structural results make the lattice refinement of the Hubbard model particularly clean. The Galerkin projection of the fine Hubbard Hamiltonian onto the prolonged subspace is itself a Hubbard model, with $(t, U) \to (t/2, U/2)$ in one dimension — so the hierarchy of resolutions is self-similar at fixed U/t — and $(t, U) \to (t/2, U/4)$ in two dimensions (Sec. III B). And the interpolation path conserves total spin and the point-group symmetry of the lattice, so the gap that controls the adiabatic transport is the distance to the next state *in the symmetry sector of the target*, which stays of order t while the sector gap of the fine Hamiltonian collapses with U and with system size (Sec. IV B); the converse is that a coarse ground state in the wrong irreducible representation cannot be refined at all (Sec. IV D).

The paper follows the structure of Ref. [1]. Section II fixes the model, the sector encoding and the ED reference. Section III defines the lattice prolongation, the Galerkin coarse Hamiltonian, the two refinement paths and the rodeo filter. Section IV reports the preliminary sector-exact results for the open chain ($2 \to 4 \to 8 \to 16$ sites) and the square plaquette ($2\times2 \to 4\times4$). Section V [will] give the Pauli structure of the Hubbard chain in the Jordan–Wigner encoding, the cost of the prolongation and of the embedded coarse term, and the gate budgets. Section VI discusses the results, the strong-coupling limit and the outlook.

II. THE HUBBARD MODEL AND ITS EXACT-DIAGONALIZATION REFERENCE

We use $t_{ij} = t = 1$ on nearest-neighbor bonds, open boundary conditions unless stated otherwise, and the lattices of Ref. [12]: the chain of L sites, the $L \times L$ square lattice with site index $i = xL + y$, and the checkerboard lattice, which adds both diagonals with amplitude αt on the plaquettes whose lower-left corner has $x + y$ even. The Hamiltonian conserves $N_\uparrow$ and $N_\downarrow$ separately, and all calculations are done in one such sector of dimension $\binom{N}{N_\uparrow}\binom{N}{N_\downarrow}$.

a. Encoding and ED. Spin-orbitals are Jordan–Wigner ordered in two blocks, the N up orbitals followed by the N down orbitals. With this ordering a down hop picks up the parity of all up electrons twice, which cancels, so the sector Hamiltonian factorizes exactly as

$$H = K_\uparrow \otimes \mathbb{1} + \mathbb{1} \otimes K_\downarrow + U\,\mathrm{diag}(\text{double occupancy}), \quad (2)$$

where K_σ is the one-species hopping matrix on the sorted occupation masks of that species. Reference energies and eigenvectors are obtained with dense diagonalization up to a few thousand states and with ARPACK Lanczos beyond; for the largest sectors the Hamiltonian is never stored, the matrix-vector product being applied as $K_\uparrow \Psi + \Psi K_\downarrow^{\mathsf{T}} + U(\mathrm{docc} \circ \Psi)$ on the state reshaped as an up×down matrix. The code is validated against free fermions at $U = 0$, the analytic dimer $E_0 = \frac{1}{2}(U - \sqrt{U^2 + 16t^2})$, the Heisenberg limit of the t–J model at half filling and the hopping-induced resonating-valence-bond states of the checkerboard lattice [12]. Every number quoted in Sec. IV is compared with this reference, and the sector dimensions encountered there range from 1 (the filled dimer) to 14 400 (four particles on sixteen sites).

b. Why the Hubbard model after the pairing model. In the pairing model the seniority symmetry confines the ground state to a subspace in which the UCCSD solver is a doubles theory and the refinement path has large gaps; the particle-hole term of Ref. [1] broke that symmetry by hand. In Eq. (1) the hopping term is a one-body particle-hole operator from the outset, single excitations do not vanish, and the only symmetries left to the pipeline are the ones it inherits from the lattice: particle number per spin, total spin S, and the point group. These are exactly the symmetries that, as we show, decide which gap governs the refinement.

III. THE PIPELINE FOR LATTICE REFINEMENT

A. Prolongation: coarse sites to blocks of fine sites

A refinement step maps a coarse lattice of N_c sites onto a fine lattice of $N_f = bN_c$ sites by resolving every coarse site c into a block $B(c)$ of b fine sites, $b = 2$ for the chain (sites $2c, 2c+1$) and $b = 4$ for the square lattice (2×2 plaquettes). The prolongation sends the coarse orbital to the normalized symmetric (bonding) combination of its block,

$$c^\dagger_{c\sigma} \;\longmapsto\; s^\dagger_{c\sigma} = \frac{1}{\sqrt{b}} \sum_{f \in B(c)} c^\dagger_{f\sigma}, \quad (3)$$

and a coarse Slater determinant to the corresponding product of $s^\dagger$ operators; particle numbers are unchanged,

so the filling halves (quarters) at each step, as it must for a sequence of lattices approaching the continuum at fixed particle number [2]. The map P is an isometry, $P^\dagger P = \mathbb{1}$, from the coarse sector of dimension $\binom{N_c}{N_\uparrow}\binom{N_c}{N_\downarrow}$ into the fine sector; its complement is spanned by configurations in which at least one antibonding block mode $a^\dagger_{c\sigma}$ (and, for $b = 4$, the two further orthogonal block modes) is occupied. On a quantum register in which the fine qubits of a block are adjacent, P is implemented by preparing the coarse state on one qubit per block per spin and applying one Givens rotation per block per spin (two for $b = 4$) — the "two single-qubit rotations and two CNOTs per site per spatial direction" of Ref. [1], i.e. $O(N_c)$ two-qubit gates, in contrast to the gate-free embedding of the basis refinement. In the sector-basis prototype used below, P is built explicitly: for each coarse occupation pattern the fine amplitudes are $b^{-n_\sigma/2}$ on every choice of one fine site per occupied block, with the sign of the permutation that sorts the chosen fine indices.

B. The Galerkin coarse Hamiltonian

The coarse Hamiltonian whose eigenstate is to be refined is the fine Hamiltonian seen at coarse resolution, $H_{\rm low} = P^\dagger H_{\rm high} P$. This choice makes the prolonged state satisfy $\langle P\psi_{\rm low}|H_{\rm high}|P\psi_{\rm low}\rangle = E_{\rm low}$ exactly, as in Ref. [1], and it has a closed form. Projecting the fine hopping term with Eq. (3): an intra-block bond contributes $-t(n_s - n_a)$, i.e. a constant $-t$ per particle on the prolonged subspace, and each inter-block bond contributes $\frac{1}{2}s^\dagger_c s_{c'}$; projecting the on-site term, $n_{f\sigma} \to n_{s_c\sigma}/b$ for every $f \in B(c)$, gives $U\sum_{f\in B(c)} n_{f\uparrow}n_{f\downarrow} \to (U/b)\, n_{s_c\uparrow}n_{s_c\downarrow}$. Hence

$$\text{chain:}\quad H_{\rm low} = H_{\rm Hubbard}\big(L;\, t/2,\, U/2\big) - tN, \tag{4}$$

$$\text{square:}\quad H_{\rm low} = H_{\rm Hubbard}\big(L\times L;\, t/2,\, U/4\big) - 2tN, \tag{5}$$

with $N = N_\uparrow + N_\downarrow$, and with the same boundary conditions as the fine lattice. Both identities were verified to machine precision against the numerically projected matrices. In one dimension the ratio U/t is preserved, so a chain of refinements $L \to 2L \to 4L \to \cdots$ is a sequence of Hubbard chains with the same interaction strength and a hopping scale that doubles at each step: the hierarchy is self-similar, and the coarsest member can be the two-site dimer or, for $N = 2L$, the filled dimer, a single computational basis state. In two dimensions the effective interaction of the coarse model is weaker by a factor of two relative to its hopping. We note that the Galerkin pair $(H_{\rm low}, H_{\rm high})$ is a choice; a continuum-scaling pair ($t \propto a^{-2}$, $U \propto a^{-1}$ in one dimension) would start the path from a different coarse ground state and change only the initial overlap p_0, not the construction.

C. Two refinement paths

a. Path A: the shift-controlled projector path. As in Ref. [1],

$$H_A(s) = \cos^2\!\big(\tfrac{\pi s}{2}\big)P(H_{\rm low}-\mu)P^\dagger + \sin^2\!\big(\tfrac{\pi s}{2}\big)(H_{\rm high}-\mu), \quad s = t/T, \tag{6}$$

with the shift $\mu = E_{\rm low} + \mu_{\rm buf}$ placing the complement of the prolonged subspace (where $P(H_{\rm low} - \mu)P^\dagger$ vanishes) at energy $\mu_{\rm buf}$ above the tracked state at $s = 0$. Because $P(H_{\rm low} - \mu)P^\dagger = \Pi_a(H_{\rm high} - \mu)\Pi_a$ with Π_a the projector onto empty antibonding modes, the coarse term is, on a device, the fine Hamiltonian in the rotated (bonding/antibonding) frame conditioned on all antibonding qubits being in $|0\rangle$ — the multi-controlled structure whose cost dominated the $3 \to 4$ chain of Ref. [1], now with $N_f/2$ control qubits per spin.

b. Path B: adiabatic un-dimerization. The same end points are connected by a path that stays two-local in the laboratory frame: add to the fine Hamiltonian an extra hopping Λ on the intra-block bonds and switch it off,

$$H_B(s) = H_{\rm high} - \Lambda\cos^2\!\big(\tfrac{\pi s}{2}\big) \sum_{\langle ff'\rangle\in{\rm intra}} \sum_\sigma \big(c^\dagger_{f\sigma}c_{f'\sigma} + {\rm h.c.}\big). \tag{7}$$

At $s = 0$ the antibonding modes are pushed up by 2Λ, and for $\Lambda \to \infty$ the ground state of $H_B(0)$ is the prolonged Galerkin ground state; at finite Λ it differs from it by an admixture of order t/Λ, so a transport that starts from the prolonged state carries a residual $1 - p \simeq c\,(t/\Lambda)^2$ to the end. Path B is the dimerized Hubbard chain (the plaquette Hubbard model in two dimensions) being adiabatically made uniform; every Trotter step along it costs what a step of the fine Hamiltonian costs, with no conditioning. We compare the two paths at equal T and, in Sec. V, at equal gate count.

c. Refinement time. The exact evolution along either path is computed as in Ref. [1]: the s interval is split into M_s slices (96 here), the Hamiltonian at the slice midpoint is diagonalized once, and the slice propagators are reused for every T. We report the overlap $p(T) = |\langle\Psi_{\rm high}|\psi(T)\rangle|^2$ with the exact fine ground state on a grid of T and quote T^* as the first grid value with $1 - p(T) < 10^{-3}$. For the 14 400-dimensional sectors the slice propagators are applied with a sparse Krylov exponential (48 slices).

D. The rodeo filter

The filter is unchanged from Ref. [1]: M cycles of the controlled evolution $e^{-iH_{\rm high}t_k}$ on one reused ancilla with the phase $e^{iE_{\rm obj}t_k}$, Gaussian cycle times of RMS width σ, and post-selection on the ancilla. For an input $|\psi\rangle = \sum_k c_k\,|E_k\rangle$ and $E_{\rm obj} = E_0$ the fidelity of the ensemble of

accepted runs obeys exactly

$$1-\mathcal{F}_M = \frac{\sum_{k\neq 0}|c_k|^2 q_k^M}{p+\sum_{k\neq 0}|c_k|^2 q_k^M}, \qquad q_k = \tfrac{1}{2}\big(1+e^{-(E_k-E_0)^2\sigma^2/2}\big), \tag{8}$$

which we evaluate from the ED spectrum. The width is set by the smallest excitation energy that carries weight in the input, $\sigma = 5/\Delta_{\rm eff}$ (Sec. IV B); choosing it from the bare sector gap would lengthen the controlled evolutions by a factor of up to three for no gain.

IV. RESULTS (PRELIMINARY, SECTOR-EXACT)

[All numbers in this section come from the sector-exact prototype and are to be regenerated, with figures, by the project notebook. The coarse state is the exact coarse ground state throughout; the gate-level UCCSD coarse solver of Ref. [1] is to be run on the $L = 4$ chain (8 qubits) and inserted as in Sec. IV.A there.]

A. The open chain refined from four to eight sites

Table I summarizes the refinement of the four-site chain with $N_\uparrow = N_\downarrow = 2$ (half filling, 36 states) to the eight-site chain (quarter filling, 784 states) for $U/t = 0, 2, 4, 8$. Three features stand out. (i) The prolonged state is a good start at every coupling, $p_0 = 0.75$–0.81, with a weak maximum at $U = 2t$; unlike the pairing model, the free-fermion case is not trivial, because the bonding modes are not the lowest fine single-particle orbitals. (ii) The sector gap of the fine Hamiltonian falls by a factor of four over the range of U, from $0.53t$ to $0.14t$, but the refinement time does not move: with $\mu_{\rm buf} = 2t$ the overlap reaches $p > 0.997$ at $T = 8/t$ and $p > 0.9999$ at $T = 16/t$ for all four couplings, and with $\mu_{\rm buf} = t$ the same values are reached at roughly twice the time. The reason is the symmetry protection discussed next. (iii) Path B refines at a comparable rate but saturates at the finite-Λ ceiling, $1 - p = 4.8 \times 10^{-3}$ ($\Lambda = 8t$) and 1.4×10^{-3} ($\Lambda = 16t$) at $U = 0$, with a weak U dependence, consistent with the $(t/\Lambda)^2$ scaling; the ceiling, not the transport, limits Path B, and a schedule with $\Lambda \to \infty$ at $s = 0$ or a final filter removes it.

B. Which gap controls the refinement: symmetry protection

Both terms of Eq. (6) commute with the total spin and with the point group of the fine lattice (the block structure respects the reflection of the chain and the C_{4v} group of the square), and the prolongation maps irreducible representations onto themselves. The transported state therefore never leaves the symmetry sector of the coarse ground state — a spin singlet, even under reflection, for the cases of Table I — and the states that can be excited along the path are only those of the same sector. The spectral decomposition of the prolonged state makes this explicit: for $U = 4t$ the lowest fine excitations at $0.223t$ and $0.459t$ above the ground state carry weights below 10^{-29}, and the first state with nonzero weight (4.6×10^{-5}) lies at $\Delta_{\rm eff} = 0.588t$; at $U = 8t$ the corresponding numbers are $0.137t$, $0.286t$ (zero weight) and $\Delta_{\rm eff} = 0.366t$ (weight 3×10^{-7}), with the bulk of the excited weight lying still higher. $\Delta_{\rm eff}$ falls with U much more slowly than Δ, which explains the U-independent T^* of Table I and the fact that the minimum of the global path gap, which sits at $s = 1$ for $\mu_{\rm buf} = 2t$ in every case, is not the relevant quantity.

The protection survives a doubling of the system: refining the eight-site chain to sixteen sites at $(2, 2)$ ($784 \to 14\,400$ states, $U = 4t$) starts from $p_0 = 0.869$, faces a fine-sector gap of only $0.051t$, and still reaches $p = 0.9993$ at $T = 16/t$ and 0.99995 at $T = 32/t$, with $E_{\rm low} = -6.8172$ and $E_{\rm high} = -7.2098$. [Compute $\Delta_{\rm eff}$ and the full $p(T)$ curve for $8 \to 16$; add the Jansen–Ruskai–Seiler bound of Sec. IV.E of Ref. [1] evaluated with $\Delta_{\rm eff}$ rather than Δ.]

C. Bootstrapping from the filled dimer

The self-similarity of the one-dimensional hierarchy allows the chain of refinements to start from the two-site dimer with $N_\uparrow = N_\downarrow = 2$, whose sector contains a single state — both sites doubly occupied — with coarse energy $E_{\rm low} = 2\cdot(U/2) - 2t\cdot 2 = U - 4t$. This is the analogue of the $k = 2$ determinant that seeded the $k = 2 \to 4$ chain in Ref. [1], and it requires no classical or variational preprocessing. Table II shows the step $2 \to 4$: the doubly-occupied seed is, as expected, a worse start as U grows, $p_0 = 0.80$ at $U = 0$ down to 0.48 at $U = 8t$, but the effective gap is large ($\Delta_{\rm eff} = 2.2t$–$0.8t$) and $p > 0.9997$ is reached by $T = 16/t$ at every U. Composed with the $4 \to 8$ step of Table I, the eight-site ground state is reached from a single computational basis state in two refinement steps of $T = 16/t$ each, with $p > 0.9999$ before any filtering. [Run the composed chain $2 \to 4 \to 8 \to 16$ with the refined (not exact) state as input of each step, as in the $3 \to 4$ chain of Ref. [1]; compare with the single-shot $2 \to 8$ and $2 \to 16$ refinements.]

D. Two dimensions: the plaquette and the wrong symmetry sector

For the square lattice the first step is $2 \times 2 \to 4 \times 4$. With one particle per spin ($16 \to 256$ states) the refinement behaves as in one dimension: $p_0 = 0.80$–0.85 for $U/t = 0, 4, 8$, $\Delta_{\rm eff} \simeq 2.1t$–$2.2t$, and $p > 0.9998$ at $T = 8/t$ (Table III). With two particles per spin ($36 \to 14\,400$ states, $U = 4t$) the prolonged coarse ground state has *zero* overlap with the fine ground state. The reason is

TABLE I. Refinement of the open Hubbard chain from $L = 4$ to $L = 8$ sites at $N_\uparrow = N_\downarrow = 2$ ($t = 1$; 36 and 784 states). $E_{\rm low}$ is the ground-state energy of the Galerkin coarse model, Eq. (4), $E_{\rm high}$ the ED ground-state energy of the fine chain, p_0 the weight of the prolonged coarse ground state on the fine ground state, Δ the sector gap of the fine Hamiltonian and $\Delta_{\rm eff}$ the lowest excitation energy carrying weight in the prolonged state. $p(T)$ is the overlap after refinement along Path A ($\mu_{\rm buf} = 2t$) and Path B ($\Lambda = 16t$); the last columns give the ensemble infidelity, Eq. (8), of the rodeo filter with $\sigma = 5/\Delta_{\rm eff}$ applied to the prolonged state and to the Path-A state refined for $T = 16/t$.

U/t	$E_{\rm low}$	$E_{\rm high}$	p_0	Δ	$\Delta_{\rm eff}$	$p_A(8)$	$p_A(16)$	$p_B(16)$	$p_B(32)$	$1-\mathcal{F}_M$, prolonged		$1-\mathcal{F}_M$, refined T=16	
										$M=2$	$M=6$	$M=2$	$M=6$
0	−6.236068	−6.822948	0.777	0.532	0.879	0.9972	0.99997	0.9966	0.9984	6.7×10^{-2}	4.5×10^{-3}	8.6×10^{-6}	5.4×10^{-7}
2	−5.437971	−6.032090	0.814	0.319	0.780	0.9988	0.99996	0.9959	0.9985	5.4×10^{-2}	3.6×10^{-3}	9.4×10^{-6}	5.9×10^{-7}
4	−4.976573	−5.634427	0.797	0.223	0.588	0.9984	0.99994	0.9932	0.9977	6.0×10^{-2}	4.0×10^{-3}	1.5×10^{-5}	9.2×10^{-7}
8	−4.558586	−5.277973	0.750	0.137	0.366	0.9976	0.99991	0.9921	0.9973	7.7×10^{-2}	5.2×10^{-3}	2.2×10^{-5}	1.4×10^{-6}

TABLE II. The first step of the one-dimensional hierarchy: the filled dimer ($N_\uparrow = N_\downarrow = 2$, one state) refined to the four-site open chain (36 states), Path A with $\mu_{\rm buf} = 2t$.

U/t	$E_{\rm low}$	$E_{\rm high}$	p_0	Δ	$\Delta_{\rm eff}$	$p(8)$	$p(16)$
0	−4	−4.4721	0.805	1.236	2.236	0.9999	1.0000
2	−2	−2.8759	0.754	0.789	1.851	0.9995	1.0000
4	0	−1.9531	0.645	0.540	1.358	0.9979	0.9999
8	4	−1.1172	0.480	0.311	0.805	0.9787	0.9997

the point group: the half-filled plaquette ground state (Galerkin parameters $t/2$, $U/4$) is the d-wave singlet that transforms as B_1 under C_{4v}, whereas the ground state of four particles on the dilute 4×4 lattice is the s-wave (A_1) singlet, and the prolongation, which commutes with C_{4v}, cannot connect them. The decomposition of the fine ground state over the prolonged coarse eigenstates confirms it: the coarse ground state carries weight 0, the first excited coarse state, only $0.071t$ above it and in A_1, carries 0.647, all other coarse states together 0.007. Refining that A_1 state along Path A ($\mu = E_1^{\rm low} + 2t$) reaches $p = 0.99997$ at $T = 16/t$ and 0.99999 at $T = 32/t$, although the fine-sector gap is only $0.090t$: once the right sector is chosen, the symmetry protection of Sec. IV B works in its favor.

Two conclusions follow for the method. The coarse state to be refined is the lowest coarse eigenstate *in the symmetry sector of the target*, which need not be the coarse ground state, and the coarse solver (UCCSD or otherwise) must therefore be run with the target's quantum numbers imposed or at least diagnosed; and the rodeo filter cannot rescue a wrong choice, since a state of zero overlap has nothing to amplify. Conversely, when the coarse ground state is in the wrong sector the refinement becomes a sensitive detector of a level crossing between the coarse and the fine problem — here between the d- and s-wave singlets as the filling goes from one half to one eighth. [Classify the irreps explicitly (build S^2 and the C_{4v} characters in the sector basis); scan U to locate the coarse B_1–A_1 crossing; compute $\Delta_{\rm eff}$ in the A_1 sector.]

E. Rodeo filtering of the refined state

With $\sigma = 5/\Delta_{\rm eff}$ ($5.7/t$ to $13.6/t$ for $U/t = 0$ to 8) the filter applied to the prolonged state behaves as the pairing case did: the infidelity falls from $1 - p_0 \simeq 0.2$ to 4–5×10^{-3} after six cycles and 2–3×10^{-4} after ten, the geometric 2^{-M} suppression of Eq. (8). Applied to the Path-A state refined for $T = 16/t$ the same filter starts from $1 - p \simeq 10^{-4}$ and reaches 0.5–1.4×10^{-6} after six cycles; applied to the Path-B state ($\Lambda = 16t$, $T = 32/t$) it starts from the ceiling, $1 - p \simeq 2\times10^{-3}$, and reaches 2–4×10^{-5} after six cycles. As in Ref. [1], the question that matters is not whether the refined input converges faster per cycle — it does, by the ratio of the initial infidelities — but whether the cycles saved pay for the refinement gates; that comparison requires the circuit-level Trotterization and is deferred to Sec. V. [Add the M-scans as a figure (prolonged, refined A at T^* and $2T^*$, refined B, UCCSD) and the acceptance probabilities.]

V. QUANTUM CIRCUITS AND DEPTH COMPARISON

[To be computed with the gate-level simulator of Ref. [1] (16 qubits for the $L = 8$ chain). What can be stated now:]

- **Pauli structure of the fine chain.** In the block Jordan–Wigner ordering of Sec. II an open chain of L_f sites on $2L_f$ qubits has $4(L_f - 1)$ hopping strings $\frac{t}{2}(X_jX_{j+1}+Y_jY_{j+1})$ of weight two, L_f interaction strings $\frac{U}{4}Z_jZ_{j+L_f}$ and $2L_f$ single-Z terms: 52 strings of weight ≤ 2 for $L_f = 8$, against the 325 strings of weight up to eight of the $k = 4$ pairing model on half the qubits. A second-order Trotter step costs of order $2\times(28+8)\times2 = 144$ CNOTs on all-to-all connectivity; the $Z_jZ_{j+L_f}$ terms are the only non-local ones and the site-major ordering, which makes them adjacent at the price of

TABLE III. Plaquette refinement $2\times2\to4\times4$ on the open square lattice, Path A with $\mu_{\rm buf}=2t$. At $(2,2)$ the coarse ground state is a B_1 singlet with zero overlap; the entry refines the lowest A_1 coarse state (see text).

$(N_\uparrow,N_\downarrow)$	U/t	dims	$E_{\rm low}$	$E_{\rm high}$	p_0	Δ	$\Delta_{\rm eff}$	$p(16)$
(1,1)	0	$16\to256$	-6.0000	-6.4721	0.805	1.000	2.236	1.0000
(1,1)	4	$16\to256$	-5.8136	-6.2638	0.840	0.792	2.096	1.0000
(1,1)	8	$16\to256$	-5.7093	-6.1833	0.845	0.711	2.129	1.0000
(2,2)	4	$36\to14\,400$	-9.4142	-10.4197	0 (B_1) / 0.647 (A_1)	0.090	[]	0.99997

weight-three hopping strings, is the alternative to be compared.

- **Prolongation.** N_c Givens rotations per spin between adjacent qubits (2 CNOTs each): 16 CNOTs for $4\to8$.
- **Coarse term of Path A.** In the rotated frame the fine Hamiltonian acquires inter-block terms coupling bonding and antibonding modes and on-site terms of weight four, and the coarse term is the bonding-mode Hubbard chain conditioned on the L_f antibonding qubits; its cost is to be counted as in Sec. V.A of Ref. [1] and is the main motivation for Path B, whose steps cost exactly a fine-Hamiltonian step plus the intra-block hopping already present.
- **Budgets.** CNOTs per accepted preparation for the prolonged, Path-A, Path-B and UCCSD inputs at matched infidelity, Trotter floor, controlled evolution of the rodeo cycles, and the sufficient conditions versus practice, all as in Secs. V.B–V.C and IV.E of Ref. [1].

VI. DISCUSSION AND OUTLOOK

[Points to make, in the order of the paper-1 discussion:]

- The lattice refinement of the Hubbard model is structurally cleaner than the basis refinement of the pairing model: a Galerkin coarse model in closed form, a self-similar hierarchy in one dimension, a two-local Hamiltonian, a prolongation of $O(N_c)$ gates. Against this stands a coarse term that is conditioned on L_f qubits, and Path B as its two-local replacement with a $(t/\Lambda)^2$ ceiling.
- Which gap matters: the symmetry-protected effective gap $\Delta_{\rm eff}$, not the sector gap; this is why T^* is flat in U and in size here, and it suggests that the inverse-gap scaling extracted in Ref. [1] through the shift should be re-read as a statement about $\Delta_{\rm eff}$. Its converse, the zero-overlap plaquette, is a warning for any refinement or nucleation scheme: the coarse state must carry the quantum numbers of the target.
- Strong coupling. At fixed U/t the hierarchy is self-similar, so refinement does not by itself reach the t–J regime; a path that raises U along with the resolution, or a refinement at fixed filling (prolonging with doubled particle number, which is not a state map) are the options, and the ED code's t–J solver gives the reference. The rodeo filter, in contrast, applies unchanged to the t–J Hamiltonian in its constrained sector. [Decide whether a t–J section belongs in this paper.]
- Relation to nucleation [11]: in one dimension the dimer-seeded hierarchy and the nucleation of a chain from a dimer start from the same state and reach the same target by refining resolution versus growing the domain; the Hubbard chain is the natural place to compare the two paths at matched gate count.
- Hardware targets: the $4\to8$ chain (16 qubits) and the $2\times2\to4\times4$ plaquette at $(1,1)$ (32 qubits, 256-dimensional sector) are within reach of current devices; the Trotter step is shallow.

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[As in Ref. [1].]

CODE AVAILABILITY

The ED code (Hubbard and t–J models, chain, square and checkerboard lattices, dense, Lanczos and matrix-free solvers) is the notebook `Hubbard.ipynb`; the sector-exact refinement and rodeo prototype used for Sec. IV is `prototype/hubbardrefine.py` with the run scripts and JSON outputs in the same directory. [Merge into the project notebook and `pairinglib`; repository link.]

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